

Experiment 2: Bridges

Objectives:

This experiment demonstrates different bridge configurations for precise resistance and impedance measurements at balance using only potentiometers. We use a DC bridge circuit to measure an unknown resistance.

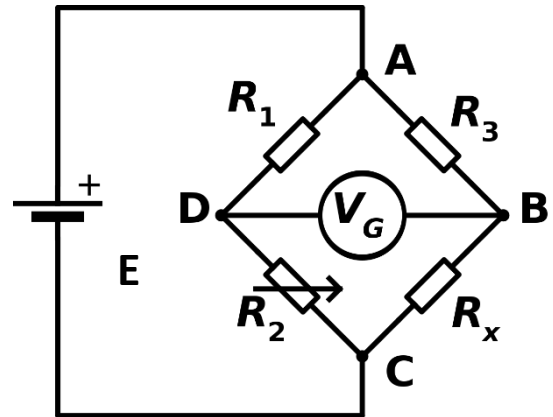
Part A: DC Bridge:

Construct the bridge with $E = 5\text{ V}$, $R_1 = R_3 = 10\text{ K}\Omega$, and R_2 variable resistor $100\text{ K}\Omega$; you can choose R_x with any value up to $100\text{ K}\Omega$ say $50\text{ K}\Omega$.

1. Measure and record the resistance of R_1 , R_3 , and R_x .

R_1	R_3	R_x

2. Change the potentiometer R_2 's value until the voltmeter reads zero voltage between the D and B points.



3. Using R_1 , R_2 , and R_3 values from 1 and 2, calculate R_x using the following equation: (prove the relation)

$$\frac{R_1}{R_2} = \frac{R_3}{R_x}$$

4. Compare between the measured R_x value in 1 and the calculated R_x value in 3.